



EE 510: Mathematical Modeling for Signal Processing

Course Code	EE 510
Course Name	Mathematical Modeling for Signal Processing
L-T-P-C	3-0-2-4
Intended for	M.Tech EE (VLSI)
Distribution	VLSI Elective
One-time approval	9th BoA (Oct 2015); 9th Senate (May 2016), in the M.Sc Applied Mathematics curriculum

Course Description

In the field of signal processing we encounter problems which requires mathematical rigor – mostly in the area of linear algebra, functional analysis, optimization and estimation theory. This course develops the mathematical framework required for signal processing which consists of certain mathematical tools as well as algorithms. How these mathematical concepts are reflected in signal processing is brought out. After finishing this course the students will be in a better position to read and appreciate the current literature and problems in this area. Course will cover Linear algebra and representation in vector spaces, Linear operator theory and matrix factorization, Estimation theory, Non-linear programming/optimization.